

Take-Home Assignment - Image Text Translation Pipeline

Context

One of our prospective clients is launching a B2B e-commerce platform in Pakistan. Their vendors upload product images (labels, packaging, promotional banners) where the text is in a foreign language. To make these products understandable to the local market, the platform needs to translate the text in each image into English and produce a version of the image with the original text replaced by the translation.

We would like you to build a working prototype of this pipeline. This is the same kind of problem you would work on in the role: take a real, slightly underspecified business need, make sensible engineering decisions, and ship something that works.

The task

Build an API endpoint that takes an input image containing text in a foreign language and returns a version of the image where that text has been replaced with its English translation, keeping the layout reasonably close to the original.

This is how the system would realistically be deployed: an endpoint the client's platform calls when a vendor uploads a product image, returning the translated version automatically. We want to see you write code in that shape.

A reasonable pipeline behind the endpoint will need to:

1. Extract the text from the image
2. Translate the extracted text into English
3. Produce an output image with the translated text in place of the original

We have deliberately left the "how" open. The interesting decisions are how you detect text, how you translate it, how you put the translation back into the image, and how you design the endpoint around all of that. Document the choices you make and why.

What we will provide

A small set of sample images (attached) for you to test against. The sample text is in Spanish. The eventual production use case will involve other languages, so please avoid hard-coding assumptions that only hold for Spanish.

Constraints and guidance

- **Time:** Aim for two to three hours. This is a prototype, not a finished product. We would rather see clean work on the common case than a sprawling attempt at every

edge case.

- **APIs and cost:** You can build this entirely on free tiers. The Gemini free tier is generous and works well for the translation step (and for text extraction if you use a vision model). Free or local OCR libraries are also completely fine. For the step that produces the output image, use whatever you judge best. If that step requires paid API usage, keep it minimal and we will reimburse reasonable costs, which should be very small.

Please use your own API keys. Setting up access to these services is part of the exercise,

and cost-aware engineering is something we value rather than penalise.

- **Use of AI tools:** You are welcome to use AI coding assistants. That is how we work. What we care about is that you understand and can explain everything in your submission.
- **No deployment or UI required.** You do not need to host the endpoint anywhere or build a frontend. You should be able to run the API locally and demonstrate it working on the sample images during the interview.
- **Scope:** Handling a single image per request is fine. Handle the common case well and write down the limitations you did not get to.

What to submit

1. **The code**, as a Git repository (a GitHub link is ideal) or a zip with the `.git` history included. We do look at how you use version control.
2. **A short README** covering: how to run the API locally, an example request, the key design decisions you made and why, known limitations, and what you would do differently with more time.
3. **The output images** your endpoint produced for the provided sample set.

Please come to the interview ready to start your API locally and walk us through it.

Optional extras (only if you have time, not expected)

- An Urdu translation in addition to English
- A simple way to check or score translation quality
- Batch processing of a folder of images
- Anything you think would make this genuinely more useful

How we will assess it

We are not grading this in isolation. The submission is the starting point for a technical conversation in the interview, where we will ask you to walk through and defend your decisions. We are looking at:

- Code quality and structure

- How you designed the API: its contract, input validation, and error handling
- The soundness of your engineering and design judgment
- How you integrated the AI components (text extraction, translation, image output)
- Whether you thought about output quality and what can go wrong
- Your use of version control and basic engineering hygiene

If anything in this brief is unclear, that is partly intentional, but you are also welcome to email us with questions. Asking good questions is a positive signal, not a negative one.

Logistics

- **Deadline:** Please submit by end of day Sunday, 24 May.
- **Interview:** Your final, in-person interview is on Monday, 25 May, at our office. We will confirm your time and share the address by email.

We are looking forward to seeing what you build.